Linear Discriminant Analysis w/ $p=1$.

Fisher 1936.

Goal: Classify observations into one of K classes ($K \geq 2$), i.e., figure out

$$p_k(x) = \mathbb{P}[Y=k \mid X=x] \quad \text{posterior probability}$$

Environment:

- π_k .. prior probability that a randomly chosen observation falls into category $k = 1..K$
- $f_k(x)$ .... density function of X for observations from class $k = 1..K$

choice, model

$f_k(x)\,dx$ ... the probability that X falls into $(x, x+dx)$ for points in class $k=1..K$

Then,

Bayes Thm

$X \in (x, x+dx)$

$$\mathbb{P}[Y=k \mid X=x] = \frac{\mathbb{P}[Y=k] \cdot \mathbb{P}[X=x \mid Y=k]}{\mathbb{P}[X=x]}$$

The Law of total Probability

Now:

$$p_k(x) = \frac{\pi_k \cdot f_k(x)}{\sum_{\ell=1}^{K} \pi_\ell \cdot f_\ell(x)}$$

$\Rightarrow$ classify into

$$k = \underset{\ell=1..K}{\operatorname{argmax}} \{p_\ell(x)\}$$

The Specifics of the LDA ($p=1$).

The Choice: f_k are normal densities for each $k = 1..K$, i.e.,

$$\boxed{f_k(x) = \frac{1}{\sigma_k\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}} \quad \text{for all } x \in \mathbb{R}$$

w/ μ_k as the mean
and
σ_k as the standard deviation for class k

Additional assumption:

Homogeneity: $\boxed{\sigma_1 = \dots = \sigma_K = \sigma}$

We now return to the posterior probabilities, i.e.,

$$p_k(x) = \frac{\pi_k \cdot f_k(x)}{\boxed{\sum_{l=1}^{K} \pi_l \cdot f_l(x)}}$$

Remember: We're looking for the k for which its $p_k(x)$ is MAXIMAL.

Since all $p_k(x)$ have the same denominator, it's sufficient to find the k such that

$$\pi_k f_k(x) \xrightarrow{k} \max$$

Because $\ln(\cdot)$ is increasing, the above is equivalent to:

$$\ln(\pi_k) + \ln(f_k(x)) \xrightarrow{k} \max$$

$$\Longleftrightarrow$$

$$\ln(\pi_k) + \ln\left(\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\frac{(x-\mu_k)^2}{\sigma^2}}\right) \xrightarrow{k} \max$$

$$\Longleftrightarrow$$

$$\ln(\pi_k) - \boxed{\ln(\sigma\sqrt{2\pi})} - \frac{(x-\mu_k)^2}{2\sigma^2} \xrightarrow{k} \max$$

constant in terms of k

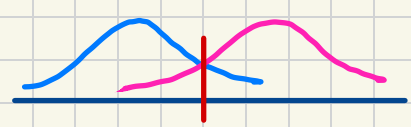
$$\Longleftrightarrow$$

$$\ln(\pi_k) \boxed{- \frac{x^2}{2\sigma^2}} + \frac{\cancel{2}x\mu_k}{\cancel{2}\sigma^2} - \frac{\mu_k^2}{2\sigma^2} \xrightarrow{k} \max$$

$$\delta_k(x) := \frac{\mu_k}{\sigma^2} \cdot x + \ln(\pi_k) - \frac{\mu_k^2}{2\sigma^2} \xrightarrow{k} \max$$

These are called DISCRIMINANT (SCORES) and they are LINEAR in x.

Special Case. $\boxed{K=2,\ \pi_1 = \pi_2 = \frac{1}{2}}$



$$\frac{\mu_k}{\cancel{\sigma^2}} \cdot x - \frac{\mu_k^2}{2\cancel{\sigma^2}} \xrightarrow{k} \max$$

$$\frac{\mu_1+\mu_2}{2}$$

$$\mu_k\left(x - \frac{\mu_k}{2}\right) \xrightarrow{k} \max$$

IF

$$\mu_1 x - \frac{1}{2}\mu_1^2 > \mu_2 x - \frac{1}{2}\mu_2^2 \text{, THEN classify as "1".}$$

$$\Longleftrightarrow$$

$$(\mu_1-\mu_2)\boxed{x} = \mu_1 \cdot x - \mu_2 \cdot x > \frac{1}{2}(\mu_1^2 - \mu_2^2) = \frac{1}{2}(\mu_1-\mu_2)(\mu_1+\mu_2)$$

Boundary is always $\frac{\mu_1+\mu_2}{2}$.

Bivariate Normal Random Pairs.

(Pitman's "Probability")

Recall: In 1D, the standard normal density is

$$\boxed{\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad \text{for all } z \in \mathbb{R}}$$

In 2D, we start w/ X and Y that are $N(0,1)$ and INDEPENDENT.

Their joint density is

$$f(x,y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} \quad \text{for all } (x,y) \in \mathbb{R}^2.$$